

HRCL: Hierarchical Relation Contrastive Learning for Low-Resource Relation Extraction

Qian Guo^{ID}, Yi Guo^{ID}, *Senior Member, IEEE*, and Jin Zhao

Abstract—Low-resource relation extraction (LRE) aims to extract the relationships between given entities from natural language sentences in low-resource application scenarios, which has been an incredibly challenging task due to the limited annotated corpora. Existing studies either leverage self-training schemes to expand the scale of labeled data, while the error accumulation of pseudo-labels' selection bias provoke the gradual drift problem in subsequent relation prediction, or utilize the instance-wise contrastive learning that fails to distinguish those sentence pairs with similar semantics. To alleviate these defects, this article introduces a novel contrastive learning framework called hierarchical relation contrastive learning (HRCL) for LRE. HRCL leverages task-related instruction description and schema-constrained as prompts to generate high-level relation representations. To enhance the efficacy of contrastive learning, we further employ hierarchical affinity propagation clustering (HiPC) to derive hierarchical signals from relational feature space with a hierarchy cross-attention (HCA) mechanism and effectively optimize pair-level relation features through relation-wise contrastive learning. Exhaustive experiments have been conducted on five public relation extraction (RE) datasets in low-resource settings. The results demonstrate the effectiveness and robustness of HRCL and outperform the current state-of-the-art (SOTA) model by 6.56% on average in terms of B³F₁. Our source code is publicly available at <https://github.com/Phevos75/HRCLRE>.

Index Terms—Hierarchical propagation clustering, low-resource relation extraction (LRE), relation contrastive learning, relational feature space.

NOMENCLATURE

X	Input text sequence.
Y	Relation label set.
H/E	Output of encoder/semantically similar instances.
$\mathcal{I}$	Instruction description.
e_i/e_s	Feature embedding of i token/structured information.
K	Number of layers of clustering.
L/p	Length of input sentence/random span.

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$h_r \in \mathbb{R}^{(2+v+p) \cdot d_R}$	Relational feature representation.
v	Length of structure information.
$f_\xi/f_{\xi'}$	Momentum/pulsion encoder.
$\mathcal{D}^k$	Demonstration example.
$\mathcal{L}$	Loss function.
Pr	Bias parameter.
$S_{n \times n}$	Initial similarity matrix.
$\hat{R}/\hat{A}$	Representative/availability matrix.
$\hat{a}_{(i,j)}/\hat{r}_{(i,j)}$	Availability/responsibility representation.
$\lambda \in [0, 1)/\varphi(i, j)$	Damping factor/sum of $\hat{a}_{(i,j)}$ and $\hat{r}_{(i,j)}$.
t/τ	Iteration step/temperature hyper-parameter.
M_l/C_l	Must-link/cannot-link representation.
α	Momentum coefficient.

I. INTRODUCTION

RELATION extraction (RE) aims to identify the relationships between given entity pairs in an unstructured text from the available relation set. For instance, in Fig. 1, reading a sentence *Glenavy was named for Glenavy, Northern Ireland, the birthplace of former New Zealand Premier John Ballance*, from the Wikipedia page, which includes a relation triplet in the form of [Head entity, Relation, Tail entity] that corresponds to the fact triplet [Glenavy, place of birth, John Ballance]. Obtaining relational fact plays a vital role in many downstream tasks of natural language processing (NLP), such as knowledge graph construction [1], information retrieval [2], and question and answering systems. While the amount of effort has been devoted to RE in supervised learning [5] that relies on manually annotated datasets, where annotated data often require a massive investment of labor and is difficult to generalize to newly relation categories in real-world application scenarios. Therefore, there is an urgent need to research low-resource RE (LRE) models that do not require large-scale annotations or very few labeled instances.

Existing LRE work can be roughly divided into two categories: data-oriented and task-oriented. The most typical method of the former is distant supervision [6], while it has an obvious flaw that easily introduces noise due to the premise that the same entity pairs share the same relation type. To fully utilize unlabeled data in low-resource settings, Rosenberg et al. [50] propose using self-training to expand the labeled dataset. However, inaccurate classification leads to error accumulation and further affects subsequent training, which is known as the gradual drift problem [10]. An example of ambiguous

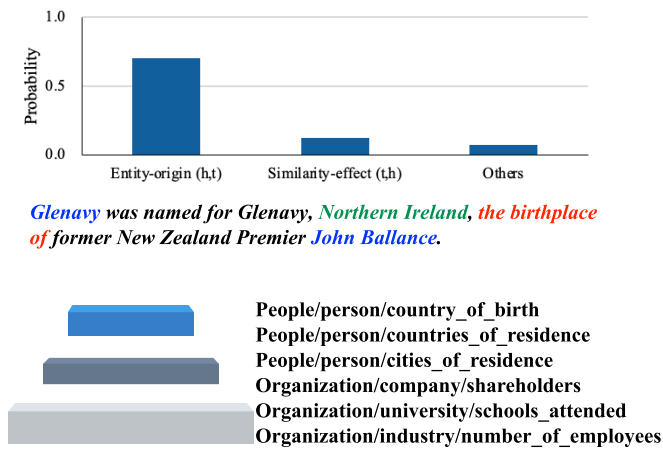


Fig. 1. Examples of confusing instances with self-training. (h, t) and (t, h) represent head and tail entities, respectively. Bottom: Relational hierarchy contained in the relational feature space, and only lists a portion of the TACRED dataset. `/people/person/country_of_birth` and `/people/person/countries_of_residence` share the same parent semantic on `/people`, meaning they belong to the same clustering exemplar from a hierarchical perspective.

instance prediction is shown in Fig. 1, where the self-training model predicts the instance as relation “Entity-origin (h, t) (Glenavy, Northern Ireland, was named for)” with a probability of 67%, as “Similarity-effect (t, h) (John Ballance, Northern Ireland, the birthplace of)” (the ground-truth label) with 24%, and as other relations with only 5%. It is hard to distinguish between the first two relations as there is a lack of understanding of hierarchical contextual semantic relations. Subsequent studies [18] propose a strategy to distinguish ambiguous but valid instances from the unlabeled set to improve pseudo-label quality. Furthermore, the iterative feedback mechanism is proposed to overcome this flaw. Hu et al. [17] correct pseudo-labels by imitating the gradient descent direction on labeled data. Later research [45] combines feature mapping with iterative feedback to improve the accuracy of pseudo-labels, which demonstrates the effectiveness of the iterative feedback method in generating correct pseudo-labels.

For the task-oriented LRE work, a common approach is to combine pre-training language models (PLMs) with few-shot learning [9], such as prompt tuning [51], which helps the model capture linguistic and semantic knowledge when performing RE or learning new relations using external knowledge [46]. For example, Chen et al. [52] propose a knowledge-aware prompt-tuning approach with synergistic optimization to enhance contextual semantics understanding. This method successfully migrates features to new relation types using only a few supporting sample examples. However, it heavily relies on task-specific prompt construction that limits the model’s generalization ability. To address this challenge, recent studies have explored contrastive learning [53]—a well-established technology in computer vision [27]. The main idea behind contrastive learning is training a model that brings positive pairs closer together in the representation space while pushing apart negative pairs. Wang et al. [24] leverage known relation classification loss to learn subtle differences between positive and negative instances and transfer the

learned knowledge to the new relation category. However, this instance-aware contrastive learning treats each sentence as an independent data point. As expanding to a larger corpus with more relation categories, the sentence pairs that share similar semantics tend to be undesirably separated in a flat relational feature space [61]. To alleviate this defect, subsequent works [11] propose exemplar-wise contrastive learning that clusters relation signals at different hierarchies, which indicates the great potential of hierarchical semantics in relation feature space for contrastive learning.

In order to alleviate the dilemma of forced separation of instances with similar semantics in contrastive learning, this article proposes hierarchical relation contrastive learning (HRCL), which adopts the idea of hierarchical cluster relational semantics of sentences based on the fact that many relation types in the corpus are organized with a hierarchical structure. For example, the relational category of the instance in Fig. 1 can be summarized as *people/country/place_of_birth* that shares the same parental semantic *people* with relation *people/country/ancestral home*, which should belong to the same category from the semantic hierarchy perspective. To this end, we attempt to explore a hierarchical cluster algorithm that incorporates propagation cluster and attention mechanism because traditional hierarchical clustering methods all suffer from the gradual drift problem. Specifically, we first define a representative instance as the clustering center in specific clustering results, denoted as the exemplar for a group of semantically similar sentences, and these exemplars can be with hierarchical structures of different granularity. To enforce relational features corresponding to the same exemplar closer than other instances at the parent granularity, HRCL encourages more reasonable clustering of relational features to obtain optimal classification results through the contrastive learning framework that integrates instance-aware and relation-aware contrastive learning.

The proposed model consists of two modules: relational representation encoder and hierarchical propagation clustering with contrastive learning. The model architecture diagram of HRCL is shown in Fig. 2. To enhance the understanding capability of PLM, we inject task-related instruction description and schema-constrained entity types together with the input text into the encoder. Furthermore, the encoder leverages the random span strategy to obtain two augmented entity-level relational features of each sentence for instance-wise contrastive learning, while the clustering algorithm retrieves hierarchical exemplars in a top-down manner for relation-wise contrastive learning and iteratively updates sentence features in a hierarchical fashion. Additionally, these updated features are further used to optimize the parameters of the encoder through the defined loss function hierarchical ReNCE (HiNCE) in this model. To summarize, the major contributions of this article are presented as follows.

- 1) We propose HRCL, a novel hierarchical propagation clustering with the contrastive learning framework for LRE that effectively suppresses the gradient drift problem in self-supervised learning through the top-down hierarchical clustering.

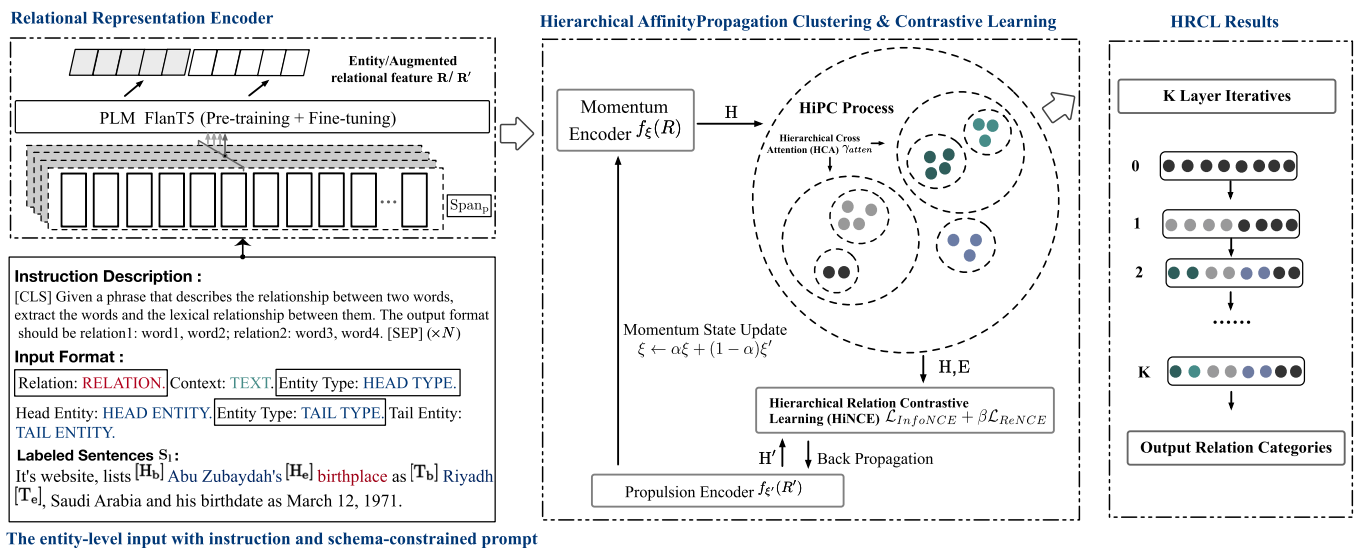


Fig. 2. Overall architecture of HRCL. Unlabeled sentences will be augmented through random spans with fixed entities, and labeled sentences include relational labels and entity types (schemas). Then, they are transmitted into the momentum and propulsion encoder with instruction descriptions, respectively. The HiPC algorithm takes momentum feature H as input and generates K layers of clustering results along with K exemplar sets E . HiNCE takes H and H' for instance-wise, while H and E for relation-wise contrastive learning.

- 2) We obtain high-level relational features by defining task-related instruction and schema-constrained entity types, which boost the semantic understanding of the encoder.
- 3) We demonstrate how to leverage the hierarchical structure of relational feature space to effectively cluster the relation instances with similar semantics via hierarchical affinity propagation clustering (HiPC).
- 4) We conduct exhaustive experiments on five available datasets, and HRCL achieves better performance than the existing state-of-the-art (SOTA) models, which strongly illustrates the superiority and robustness of the proposed model for LRE by leveraging relation-wise contrastive learning. The ablation studies also demonstrate the crucial role of individual modules within our framework.

The remaining of this article is organized as follows. Section II surveys the related work with contrastive learning and LRE. Section III gives the problem definition and executive details of the proposed model, and the experiments are presented in Section IV. Finally, we conclude this article and give an outlook on future research in Section V. Note that Section II is the theoretical basis and provides theoretical support for Sections III and IV. Sections III and IV are the essential portions of this work.

II. RELATED WORK

A. Low-Resource Relation Extraction

The scarcity of labeled data inspires the fact that LRE has received more attention. There are two major application scenarios in LRE: low-shot RE and low-resource supervised RE. Low-shot RE, including few-shot RE [3] and zero-shot RE [54], trains a model to recognize unseen relation labels in a test set with a few support instances or even none of the instances. The low-resource supervised RE aims to train a

supervised model with fewer training data [55]. In this work, we focus on the low-resource supervised RE.

The LRE model is mainly divided into two directions, one is to design a robust model architecture that fully utilizes very few annotated data to obtain relationship semantics. For instance, Dong et al. [8] incorporate label semantics into the pre-training and fine-tuning model with a few labeled data. Meanwhile, Zhang et al. [16] proposed fully exploring the explicit relational semantics in relation labels based on an external knowledge graph. Another direction is to expand annotation data through the self-training approach automatically [18], [56]. A typical approach [17] is assigning pseudo-labels by exploiting a contextualized augmentation method to generate data.

The above methods have drawbacks in task generalization and model complexity or suffer from the gradient drift problem due to the noisy pseudo-labels without considering the high-level relational features. In this article, we leverage PLM to extract high-quality relational features by integrating rich semantic knowledge. Furthermore, these semantic features enable the proposed relation-wise contrastive learning framework to perform relation clustering efficiently.

B. Contrastive Learning

The main challenge in LRE is obtaining key semantic features from limited annotated data. An important milestone [20] is the self-supervised learning strategy, assuming the cluster assignments as pseudo-labels and a classification objective is optimized. Soares et al. [12] propose continuously updating the context representation for supervised learning by assigning pseudo-labels to the clustering center and regarding the RE task as the optimization classification goal. Although these methods can achieve influential clustering effects from contextual relation features, they have certain defects that interfere with representation learning due to initialization

requirements in the bottom layer. He et al. [27] propose online deep clustering, which conducts clustering and network updates simultaneously rather than alternating to tackle these clustering points. Maddouri et al. [62] propose geometric-AP, a novel clustering algorithm that effectively extends affinity propagation [63] to take advantage of the network topology. Zhou et al. [9] propose a multicenter cluster contrastive loss that enables a relation type corresponding to multiple cluster centers to match the pre-training representation better. Later studies [22] conducted hierarchical clustering to extract semantic relation wrapped in the context by discovering the hierarchical structure. However, these works heavily rely on the hierarchical structure in the context, which results in clustering model instability.

Inspired by the success of contrastive learning in computer vision tasks, instance-wise contrastive learning [23], [24], [60] is receiving more and more attention. Zhou et al. [9] propose a pre-train and finetune model using consistent objectives of contrastive learning for LRE and largely outperforms PLM-based RE classifier. Li et al. [21] propose a hierarchical contrastive learning framework (HiCLRE) to decrease data noise by fusing global structural knowledge and local fine-grained interaction. Hu et al. [11] explore an exemplar-wise contrastive learning model that leverages the hierarchical information in relation feature space to cluster semantic similar instances. Li et al. [58] study the sentence- and pair-level contrastive learning objectives to differentiate between different classes of sentence pairs by bringing those in one class together and pushing those in another class apart. Peng et al. [59] propose the dual curriculum contrastive module (DCCM) to decrease the redundant information of learned latent variables for graph clustering.

However, the above methods cannot learn high-level relation representations in low-resource settings. In our work, we focus on using relation-wise contrastive learning for LRE. It has the superiority of supervised learning to get high-quality semantic features with fewer annotated data instead of exploiting base-level sentence differences to boost discriminative capability. Also, it retains the advantage of unsupervised learning to deal with cases where there are a large number of new relations in samples.

III. MODEL METHODOLOGY

A. Model Overview

As illustrated in Fig. 2, the proposed model includes two components: a relational representation encoder and hierarchical propagation clustering with relation contrastive learning. The encoder takes natural language sentences as input, which include instruction description, entity types, and relational categories as prompts, together with the context text, then employs PLM FlanT5 fine-tune to output two contextualized representation sets R and R' for the purpose of contrastive learning.

Afterward, the clustering module takes these feature embeddings to retrieve exemplars (relation categories) that represent a group of semantically similar instances in the hierarchical structure, referred to as E . Subsequently, these examples are

employed to iteratively update relational features of instances in a hierarchy with relation-wise contrastive loss ReNCE that enforces the relation feature to be aligned with its corresponding cluster center. The frequently used notations in this article are summarized in Nomenclature for quick reference.

B. Relational Representation Encoder

The relational representation encoder aims to obtain two relational features from each instance based on the structural prompt and relational category for instance-wise contrastive learning.

For a sentence $X = [x_1, x_2, \dots, x_l]$ with l words, where x_i denotes a word and Head and Tail entity mentions, we utilize four reserved tokens to argument X representation. Specifically, token $[H_b]$, $[H_e]$, $[T_b]$, and $[T_e]$ are introduced to represent the beginning and end position of each entity mention in the input text. These special tokens are injected into X , denoted as follows:

$$X = [x_1, \dots, [H_b], \dots, x_i, \dots, [H_e], \dots, x_{\text{span}1}, \dots, x_{\text{span}p}, \dots, [T_b], \dots, x_j, \dots, [T_e], \dots, x_l]. \quad (1)$$

After that, X together with relation y_i , entity type e_i and instruction description $\mathcal{I}$ defined as a tuple $X' = \{ \langle X; y_i; e_i; \mathcal{I}_j \rangle \mid y_i \in Y \}$ input into the context encoder, where Y denotes the pre-defined relation set. Then, X' will be the input sequence for the context encoder to extract relational features, where $x_{\text{span}i}$ indicates the random span words. In view of the powerful language representation ability of PLM, we utilize FlanT5 [49] to effectively encode each token in the input sequence along with context and further fine-tune with the specific structure information to obtain the feature embedding e_i , where $i \in [1, L]$, including the structured information in x' and $e_i \in \mathbb{R}^{d_R}$, where d_R represents the dimension of the token embedding.

Furthermore, for contrastive learning purposes, we randomly select p words as Random Span from all the context words in the whole sentence to augment the relational features as e_{span} , where multiple different Random Span selections result in different semantically embedding invariant of the same sentence. For every selection, we concatenate the structure information and relation entity embedding and p Random Span words together to derive a fixed-length relational feature $h_r \in \mathbb{R}^{(2+v+p) \cdot d_R}$

$$h_r = [e_s \oplus e_{\text{entity}1} \oplus e_{\text{entity}2} \oplus e_{\text{span}1} \oplus \dots \oplus e_{\text{span}p}] \quad (2)$$

where $\oplus$ denotes the concatenation operator, e_s represents the structured information embedding, m is the length of structure information, h_r is the output of the relation representation encoder which can be denoted as $f_{\xi}(x', I_s, \text{Entity}_h, \text{Entity}_t, x_{\text{span}})$. The Random Span strategy is an effective data augmentation method that can get sentence-level relational features to construct positive samples directly.

C. Hierarchical Propagation Clustering With Contrastive Learning

To increase the information content of positive samples during the contrastive learning phase, we designed a hierarchical

propagation clustering algorithm. This algorithm is based on limited prior knowledge and produces multilayer clustering results that aggregate instances with similar semantics as positive samples.

1) *Improved Hierarchical Propagation Clustering*: In order to obtain hierarchical structure information between data points, we propose an HiPC method based on the affinity propagation (AP) algorithm. The AP algorithm differs from traditional clustering methods such as k -means. It considers each data point as a node in the graph and selects the optimal set of relation representative points (exemplars) by propagating information within the graph. However, traditional clustering methods improve clustering quality through an iterative process that continuously updates cluster centers. This approach disregards hierarchical information in the dataset and necessitates predetermined cluster numbers. In addition, the AP algorithm propagates using similarity information among data points to obtain optimal class representative points for each relation and optimize the objective function. It treats all data points as candidate class representative points, thus avoiding limitations in clustering results caused by the selection of initial exemplars. Furthermore, it neither requires the actual number of target relations in advance nor the distribution of relational type.

a) *Affinity propagation clustering*: Assuming that there are n data points in clustering space, the initial similarity matrix s is denoted as $S_{n \times n}$. Affinity propagation clustering (APC) needs to set the value of the bias parameter Pr for each x_i . Generally, the initial assumption is that Pr with the same value p' . Furthermore, two important parameters are defined as the representative matrix $\hat{R} = [\hat{r}(i, j)]_{n \times n}$ and the availability matrix $\hat{A} = [\hat{a}(i, j)]_{n \times n}$, respectively. Responsibility $\hat{r}(i, j)$ points from point x_i to point x_j is the accumulation evidence of x_j , indicating the appropriateness of x_j to be the class representative point of x_i , while the availability $\hat{a}(i, j)$ points from point x_j to point x_i is the cumulative degree of x_i , which indicates the appropriateness of x_i to choose x_j as its exemplar. The class representative point is denoted by $\arg\max_j (\hat{r}(i, j) + \hat{a}(i, j))$. Besides, a damping factor λ is introduced to prevent model from oscillating. The calculation formula for whole process is as follows:

$$\hat{r}(i, j) \leftarrow s(i, j) - \max_{j' s.t. j' \neq j} \{\hat{a}(i, j') + s(i, j')\} \quad (3)$$

$$\text{if } i \neq j, \hat{a}(i, j') \leftarrow \min\{0, \hat{r}(j, j) + \sum_{i' s.t. i' \notin \{i, j\}} \max\{0, \hat{r}(i', j)\}\} \quad (4)$$

$$\hat{a}(j, j) \leftarrow \sum_{i' s.t. i' \neq j} \max\{0, \hat{r}(i', j)\}. \quad (5)$$

b) *Hierarchical affinity propagation clustering*: To further improve the clustering performance, HiPC is proposed by defining a new similarity matrix s' with known label. Generally, there are two types of pairwise constraints [26] Must-link and Cannot-link: Must-link indicates that two points must belong to the same class, namely $(M_l = \{(x_i, x_j)\})$, while Cannot-link specifies that two points are not in the same class, that is, $(C_l = \{(x_i, x_j)\})$. Thus, the labeled data is converted

into pairwise constraints for calculation, and the adjusting of the similarity matrix is defined as follows.

- 1) When pairwise constraints $\{(x_i, x_j)\} \in M_l$, the highest similarity possible value is 0. Therefore, if the pairwise constraints belong to the same class, they should be adjusted accordingly with $s'(i, j) = 0$.
- 2) When the pairwise constraints $\{(x_i, x_j)\} \in C_l$, indicating that there is a very low similarity between two data points, the minimum similarity is defined as $-\infty$ by the similarity measure.

The specific adjustment steps for the similarity matrix are as follows.

- 1) The current existing Must-link data pairs satisfy the condition: $(x_i, x_j) \in M_l \Rightarrow s(i, j) = 0 \& s(j, i) = 0$. Extending the known Must-link pairwise constraints to a new one and adding the new Must-link pairwise constraint to the Must-link constraint set: $(x_i, x_o) \notin M_l \& (x_i, x_j) \in M_l \& (x_j, x_o) \in M_l \Rightarrow s'(i, o) = 0 \& s'(o, i) = 0 \& M_l = (x_i, x_o) \cup M_l$.
- 2) Adjusting the similarity of data pairs that satisfy the Cannot-link constraint in the prior knowledge: $(x_i, x_j) \in C_l \Rightarrow s'(i, j) = -\infty \& s'(j, i) = -\infty$.
- 3) Globally adjusting the similarity of data pairs not contained in the prior information again. The formula is expressed as: $(x_i, x_j) \notin \{M_l \cup C_l\} \Rightarrow s'(i, j) = \max\{s(i, j), s(i, o) + s(o, j)\}$.

Since only the data point x_o in the Must-link set can alter the value of $s'(i, j)$, we compare its calculation solely with the corresponding data point in a specific pairwise constraint of M_l set to expedite computation. Algorithm 1 outlines all calculation steps.

Algorithm 1 HiPC

Input: Clustering layer M , Hidden vector H .

Output: Clustering results C .

```

1 Initialize: Initilize  $\{s'_{i,j} | (i, j) \in [1, n]\}$ 
2  $\forall i \neq j, p_u = \text{med}(s_{ij}), p_l = \text{mean}(s_{ij}),$ 
3  $p' = \{P_r | P_r = p_u + \frac{p_u - p_l}{M-1}(m-1), m \in [1, M]\}$ 
4 while ( $m < M$  or  $\Delta I < \text{threshold}$ ) do
5   Update  $s_{ij}$  according to  $H^m$ , Set  $s_{ii} = p$ .
6   for ( $i=1; i < n; i++$ ) do
7     Update information  $\tilde{c}_i \leftarrow \arg\max_j (\bar{a}_{i,j} + \bar{r}_{i,j}),$ 
8      $e(h_i, h_j) \leftarrow \varphi(h_i, h_j) - \max_{j' s.t. j' \neq j} \{\varphi(h_i, h_{j'})\},$ 
9      $E^m = \{e_{\tilde{c}_i}^m | e_{\tilde{c}_i}^m = h_{\tilde{c}_i}^m, \tilde{c}_i \in \tilde{C}\}.$ 
10    if ( $\text{len}(C)$  no changes) then
11      Break;
12    Update the  $p$  value.
13   $C.add(E^m), H^{m+1} \leftarrow (H^m, E^m)$  by HCA.
14 return  $C$ .
```

As shown in Algorithm 1, through two forms of data pairwise constraints, we hope to discover the clustering results of M consecutive layers, where the data points clustering in the m layer is close to the corresponding clustering center

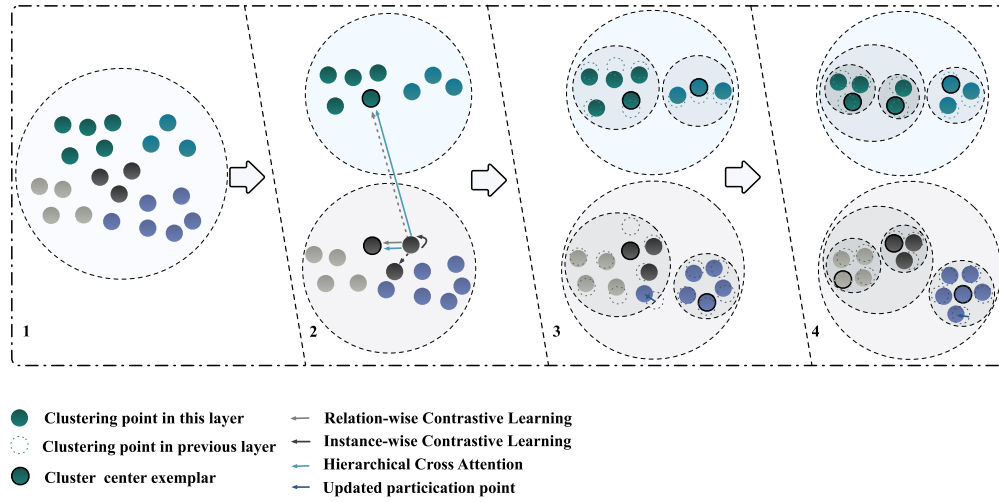


Fig. 3. Information propagation process of clustering with HCA. The first part indicates the original data points to be clustered. The second part divides data points into two cluster centers, and the cross-attention mechanism iteratively updates information in the different levels, which assists in the next level of clustering. The solid and dotted arrows represent the positive and negative instance pairs and clustering direction in contrastive learning. The dotted circle represents the clustering point from the previous layer, gradually approaching the exemplar at the clustering center in the third part. The fourth part displays the final clustering result.

exemplar of the $m - 1$ layer, and the number of clustering results is determined by the preference setting Pr .

2) *Hierarchy Cross-Attention*: Traditional hierarchical clustering either divides coarse-grained classes into fine-grained ones in a top-down manner or merges fine-grained classes into a coarse-grained tree structure in a bottom-up manner. However, these methods do not consider the information interaction between adjacent layers, which can lead to an errors accumulating over time. To address this problem, we propose using the hierarchy cross-attention (HCA) algorithm to facilitate information interaction between coarse-grained and fine-grained levels during the clustering stage.

To derive the concern between points at position (j, o) , we calculate the attention weight matrix A^m using class representative point e_i in the m th clustering layer

$$a_{jo}^m = \frac{\exp(\omega e_j^m \cdot e_o^m)}{\sum_{o'} \exp(\omega e_j^m \cdot e_{o'}^m)} \quad (6)$$

where ω denotes the trainable scalar variable. The attention weight a_{jo}^m aggregates the interaction information between these exemplars at the (j, o) position at layer m , which directly reflects the degree of approximation between the data points corresponding to the exemplars. Generally, when class representative points are closer together visually, their attention weight will be higher. By summing the attention weight with its corresponding exemplar from other exemplars, we obtain an attended point representation at layer $m + 1$

$$\tilde{h}_i^{m+1} = \sum_o a_{jo}^m e_o^m \quad (7)$$

where $\tilde{h}_i^{m+1} \in \mathbb{R}^d$, e_o^m represents the exemplar of h_i^m , namely the cluster center. The attended point representations aggregate the most associated information and propagate them from the m to $m + 1$ layer according to how close they are to exemplar e_o^m . They reflect the correlation of clustering

information between adjacent layers and how to get close to these data points with different categories. Based on this approach of information propagation, we define the embedding representation h_i^{m+1} of HCA by combining the attended point representation with the original one as follows:

$$h_i^{m+1} = h_i^m + \omega_{\text{att}} \tilde{h}_i^{m+1} \quad (8)$$

where ω_{att} is the automatically trained weight variable. HCA effectively alleviates the information shielding problem between adjacent layers and enforces data points to get closer to corresponding exemplars in the previous layer. Thus, it can achieve better clustering results in the current layer. The clustering process between data points is dynamically illustrated in Fig. 3.

3) *Hierarchical Relation Contrastive Loss*: Given the training set $D = \{d_1, d_2, \dots, d_n\}$ that contains n sentences, the relational representation encoder outputs relational feature and its augmented view for each input sentences by randomly sampling spans twice, denoted as R and R' . After performing the random span selection for all sentences, obtaining two embedding representations $R = \{r_1, r_2, \dots, r_n\}$ and $R' = \{r'_1, r'_2, \dots, r'_n\}$, where $R, R' \in \mathbb{R}^{n \times l \times d}$. The contrastive learning model aims at dispersing differences between a batch of instances to better represent relational features. It iteratively updates the amount of information by optimizing a contrastive loss objective InfoNCE [4]

$$\mathcal{L}_{\text{InfoNCE}} = - \sum_{i=1}^n \log \frac{\exp(r_i \cdot r'_i / \tau)}{\sum_{j=1}^L \exp(r_i \cdot r'_j / \tau)} \quad (9)$$

where $r_i \in R$ and $r'_i \in R'$ are positive sample representations for instance i , while r'_j includes one positive sample and $L - 1$ negative samples for other instances, and τ represents the temperature hyperparameter.

Compared with the instance-wise contrastive learning that takes all data pairs beyond the same relational category as

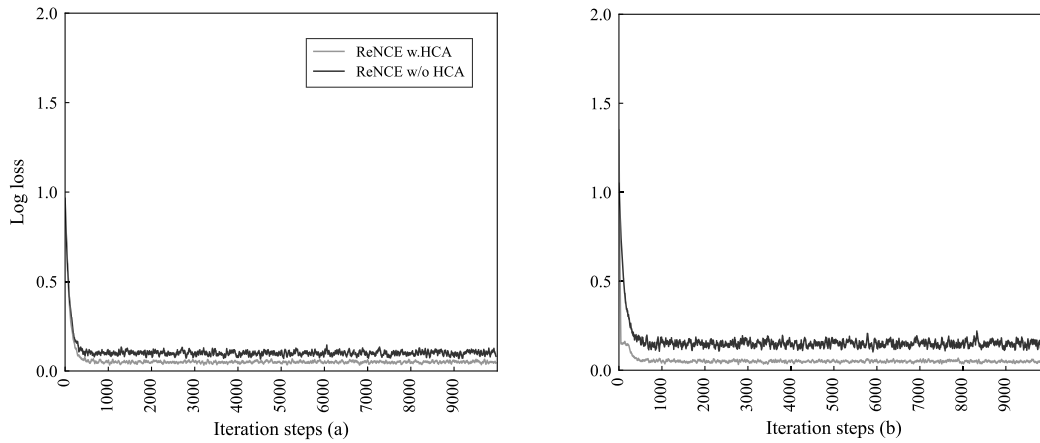


Fig. 4. Effect of HCA on (a) TACRED dataset and (b) FewRel dataset. Recording the average value during training for the first 10 000 iteration steps across five runs.

the negative samples while ignoring the semantic similarity between them, we leverage the hierarchical semantic structure in the relational feature space to achieve a better aggregation of different relational categories with similar semantics. The HiPC will continuously perform information iteration by utilizing input relational features in each epoch, and the implement process is illustrated in Fig. 2. It is worth noting that the relational feature h_i^m will be updated through HCA in each batch size during the training phase, while the cluster center will not be retrieved until the iteration process of each epoch is completed. In order to prevent the clustering center exemplar from drifting and keep the smooth transition of extracted relation features, we employ the Momentum Encoder $f_\xi()$ [27] and Propulsion Encoder $f_{\xi'}()$ to ensure a more smooth update during the propagation clustering.

Both Momentum and Propulsion Encoder is an instance of the relational representation encoder, where the contrastive learning loss is used to update ξ' , and ξ is a moving average of the updated ξ' to ensure a stable update of relational features. The Momentum Encoder outputs relational feature $h_i = f_\xi(r_i)$ to the HiPC module and conducts propagation cluster to obtain clustering center set C , which includes the K layers clustering result c_k . In order to achieve a better clustering effect, that is, to enforce more relational features to be closer to the corresponding exemplar, we jointly optimize the loss function through relational contrastive learning, and define the relational contrastive loss as ReNCE

$$\mathcal{R}_{\text{ReNCE}} = -\frac{1}{N} \sum_{i=1}^N \frac{1}{K} \sum_{k=1}^K \log \frac{\exp(h_i \cdot e_j^k / \tau)}{\sum_{v=1}^{c_k} \exp(h_i \cdot e_v^k / \tau)} \quad (10)$$

where $j \in [1, c_k]$ and e_j^k is the corresponding exemplar of point i at k layer, and N is the number of instances. Note that h_i and e_j^k are constrained in the approximate feature space, the temperature parameter τ is shared with $\mathcal{L}_{\text{InfoNCE}}$ here. It can be seen from the second part of Fig. 3 the difference between $\mathcal{L}_{\text{ReNCE}}$ and $\mathcal{L}_{\text{InfoNCE}}$, where the positive and negative instances of the same category and different categories are represented by distinguishing between solid and dashed arrows. Moreover, to stabilize the information iteration in the process of propagating clustering, we also retain $\mathcal{L}_{\text{InfoNCE}}$ to jointly optimize

TABLE I

STATISTICS OF FIVE DATASETS. 30/10/143 DENOTES THE NUMBER OF INSTANCES CORRESPONDING TO EACH RELATION WITH A RANDOM SAMPLE FROM THE TRAINING, VALIDATION, AND TESTING SET

Datasets	Instances	Entities	Relations	Instances per Relation
SemEval	10,717	7,984	19	30/10/143
TACRED	106,264	—	41	20/10/145
Re-TACRED	91,467	—	40	20/10/145
FewRel	56,000	72,954	80	20/10/100
SciERC	4,716	8,089	7	50/20/20

the contrastive loss objective, and the final loss function is expressed as follows:

$$\mathcal{L}_{\text{HiNCE}} = \mathcal{L}_{\text{InfoNCE}} + \beta \mathcal{L}_{\text{ReNCE}}. \quad (11)$$

The hyperparameter β is utilized to balance the two optimized objective functions. Once the Propulsion Encoder $f_{\xi'}$ has been updated with $\mathcal{L}_{\text{ReNCE}}$, the momentum updated rule for parameter ξ can be derived as follows:

$$\xi \leftarrow \alpha \xi + (1 - \alpha) \xi' \quad (12)$$

where the constant $\alpha \in [0, 1]$ is a momentum coefficient, which usually takes the value close to 1 for a smoother transiting.

IV. EXPERIMENTS AND ANALYSIS

A. Experimental Setup

1) *Real Benchmark Datasets*: We conduct experiments on five reconstructed benchmark datasets, including the SemEval2010 Task8 (SemEval) [28], TAC RE Dataset (TACRED) [29], Re-TACRED [44], FewRel [3], and SciERC [43] to evaluate the performance of HRCL. Table I illustrates the statistical details of these datasets.

1) SemEval 2010 Task8 contains nine bidirectional relation categories, where the *no_relation* is included and accounts for 17.4%. Besides, there are 10717 instances in total in the dataset with unequal relational category. Specifically, there are 7199 relation mentions in the training set, and 800 and 1864 in the validation and testing sets, respectively. To satisfy the low-resource

setting, we randomly select 30 instances for each relation from the original training set as manually annotated samples and the remaining instances as unlabeled samples. Moreover, we also reconstruct the validation set by randomly sampling ten instances for each relation from the original validation set. Therefore, there are 570, 190, 2717, and 5870 instances for the training set, validation set, testing set, and unlabeled data.

- 2) TACRED is a large-scale crowd-sourced RE dataset that contains 42 relations (including *no_relation*), and the percentage is 78.7% and 106 264 instances, which is collected from online documents and news media. There are 75 049, 25 763, and 18 659 relation mentions in training, validation and testing set, respectively. Besides, due to the unbalanced distribution of relational instances, we have excluded the special relation *no_relation*. Specifically, we limit the number of training instances to 20, validating instances to 10 for each relation from the original training and validating set, and the remaining instances as the unlabeled data. Therefore, the number of instances in the training, validation, testing and unlabeled sets is 780, 390, 5648, and 18 938, respectively.
- 3) Re-TACRED is the repaired version of TACRED, which contains 91 467 sentences encompassing 41 relations (including *no_relation*). We randomly sample 20 instances for each relation from the original training set, ten instances from the original validation set, and utilize the remaining instances as unlabeled data for testing. Furthermore, we exclude the special “*no_relation*” type as it accounts for over 60% of the instances.
- 4) FewRel sourcing from Wikipedia articles by distantly aligning WikiData triplets, which is widely used in LRE tasks. It contains 100 relational categories that correspond to 70 000 instances. We randomly sample 20/10/100 instances for each relation type from the original training, validation, and test set to meet the low-resource setting.
- 5) SciERC collects data from 12 AI conference/workshop sessions organized by four AI groups, holding 500 scientific abstracts annotated with coreference clusters, relations, and academic entities.

2) *Compared Methods*: For baselines, we compared HRCL with five representative SOTA methods and categorized them into three groups, which include: 1) with self-training methods fmLRE and STAD; 2) clustering method of contrastive learning MCCL and MatchPrompt; and 3) GPT-3 baselines GPT-Sent.

- 1) fmLRE [45] based on feature mapping similarity calculation to improve the quality of pseudo-labels with an iterative feedback mechanism. We perform the comparative experiment directly on the datasets of low-resource settings without modifying the model’s original parameters.
- 2) STAD [18] adapts a joint-training method, building the RE system on confident and uncertain instances, improving the performance of LRE. Here, we test STAD

on four datasets of low-resource settings and report the best performance.

- 3) MCCL [9] employs a multicenter contrastive loss that allows one relation to form multiple clusters to better align with pretraining. Note that MCCL only considers two biomedical literature-related datasets, so we adjust the relevant parameters of MCCL, test its performance on the low-resource dataset, and report the best performance.
- 4) MatchPrompt [46] is an open RE framework that excavates the specific meanings for cluster interpretability and efficient knowledge from pre-defined relational instances. We evaluate its efficiency on four low-resource datasets and report the best performance to compare with HRCL.
- 5) *GPT-Sent*: For the GPT-3 baseline, we select “text-davinci-003” [48] and adopted Sim-CSE [47] for sentence embedding as it has been demonstrated to be the SOTA method for sentence similarity tasks. In our experiment, we utilize PLM version: sup-simcse-bert-large-uncased.

3) *Implementation Details*: We conducted experiments using the Pytorch framework [35] and FlanT5 [49] as the underlying encoder. During the encoder phase, we set the random span value p to 3, which is explained in detail in Section IV-B3. Therefore, the output relation representation $r_i, r'_i \in \mathbb{R}^{5 \times d_r}$, where $d_r = 768$. Additionally, we initialized the Momentum Encoder f_ξ and Propulsion Encoder $f_{\xi'}$ with FlanT5. The encoder was trained for 30 epochs using AdamW [36] optimization with a learning rate of $2e^{-5}$ and weight decay of 0.01. In the HiPC phase, we set the number of clustering layers K is 10 and the maximum iterations to 500 to ensure the model terminates timely, and the convergence condition of the clustering algorithm is E^m will not change for 20 iterations. The temperature parameter τ is set $\{0.1, 0.02, 0.05, 0.04, 0.06\}$ for SemEval 2010, TACRED, FewRel, Re-TACRED and SciERC, respectively. We follow He et al. [27] setting the momentum coefficient and readjusting the number of negative samples $\mathcal{N}$ to 384 to satisfy the requirements of small batch size. All experiments were implemented on three NVIDIA RTX A6000 GPUs with a memory capacity of 48 GB.

4) *Evaluation Metrics*: We follow the previous work [37] and adopt three general metrics B^3 [38], V-measures [39], and adjusted rand index (ARI) [40] to measure the clustering effectiveness. For B^3 , measuring the correct rate of its cluster or clustering all instance points into different clusters by using precision and recall. Then, we calculate the harmonic mean B^3F_1 of the B^3 precision and recall as follows:

$$B^3\text{Precision} = \exists_{X,Y} P(g(X) = g(Y)|e(X) = e(Y))$$

$$B^3\text{Recall} = \exists_{X,Y} P(e(X) = e(Y)|g(X) = g(Y)) \quad (13)$$

where X, Y corresponding to a random variable of input instance, $e(X)$ denotes the predicted cluster center of X , and $g(X)$ is the truth gold class. Besides, we also report the F_1 and accuracy values, which indicate the accuracy rate of correctly

TABLE II

PERFORMANCE OF DIFFERENT MODELS IN TERMS OF B^3 , V-MEASURE, ARI, AND ACC (%) ON THREE DATASETS. THE NUMBERS BEHIND THE “ $\pm$ ” SIGN REFER TO ITS STANDARD DEVIATION. METHODS MARKED WITH * ARE THE COPIED RESULTS DIRECTLY FROM THE ORIGINAL LITERATURE. METHODS MARKED WITH $\dagger$ REPRESENT THAT WE RE-IMPLEMENT THE MODELS FOR THIS TASK

Dataset	Models	B^3			V-measure			ARI	Acc(%)
		F_1	Prec.	Rec.	F_1	Hom.	Comp.		
SemEval	STAD*	77.30 $\pm$ 1.40	75.23 $\pm$ 1.31	79.49 $\pm$ 1.57	79.24 $\pm$ 1.35	70.28 $\pm$ 2.03	90.82 $\pm$ 2.21	53.24 $\pm$ 2.05	65.38
	fmLRE*	<u>85.97</u> $\pm$ 1.32	79.38 $\pm$ 1.59	93.75 $\pm$ 2.73	<u>86.49</u> $\pm$ 1.41	80.32 $\pm$ 1.57	<u>93.68</u> $\pm$ 1.23	<u>65.73</u> $\pm$ 1.34	<u>79.67</u>
	MCCL $\dagger$	78.24 $\pm$ 2.26	78.43 $\pm$ 2.09	78.05 $\pm$ 2.17	78.68 $\pm$ 2.03	75.17 $\pm$ 1.90	82.53 $\pm$ 2.14	59.28 $\pm$ 1.45	65.29
	MatchPrompt $\dagger$	75.43 $\pm$ 1.31	69.84 $\pm$ 1.73	81.99 $\pm$ 1.75	77.34 $\pm$ 1.81	71.23 $\pm$ 2.13	84.59 $\pm$ 1.25	54.36 $\pm$ 2.16	63.45
	HRCL(Our model)	89.47 $\pm$ 1.21	90.14 $\pm$ 1.03	<u>88.81</u> $\pm$ 1.56	87.13 $\pm$ 1.04	<u>79.37</u> $\pm$ 0.79	96.57 $\pm$ 0.89	68.39 $\pm$ 0.96	80.19
	GPT-Sen $\dagger$	78.85 $\pm$ 1.53	<u>80.10</u> $\pm$ 1.45	77.63 $\pm$ 2.04	75.62 $\pm$ 1.79	76.48 $\pm$ 1.69	74.77 $\pm$ 2.03	56.49 $\pm$ 1.78	64.57
TACRED	STAD $\dagger$	60.39 $\pm$ 1.47	63.24 $\pm$ 1.37	57.78 $\pm$ 1.54	62.47 $\pm$ 1.39	65.48 $\pm$ 2.01	59.72 $\pm$ 2.54	64.27 $\pm$ 2.54	69.43
	fmLRE*	65.71 $\pm$ 1.78	66.83 $\pm$ 1.47	64.62 $\pm$ 1.56	69.26 $\pm$ 1.59	70.17 $\pm$ 1.04	68.37 $\pm$ 1.46	67.28 $\pm$ 2.13	70.14
	MCCL $\dagger$	61.39 $\pm$ 1.38	64.74 $\pm$ 1.59	58.37 $\pm$ 2.19	59.27 $\pm$ 2.46	61.74 $\pm$ 1.83	56.99 $\pm$ 2.34	62.71 $\pm$ 1.75	66.52
	MatchPrompt*	<u>87.43</u> $\pm$ 1.09	86.62 $\pm$ 1.37	<u>88.26</u> $\pm$ 0.79	<u>89.30</u> $\pm$ 1.35	89.01 $\pm$ 1.51	<u>89.73</u> $\pm$ 1.69	<u>83.20</u> $\pm$ 1.87	<u>75.29</u>
	GPT-Sen $\dagger$	77.69 $\pm$ 2.02	73.48 $\pm$ 1.24	82.41 $\pm$ 1.84	79.83 $\pm$ 1.67	75.49 $\pm$ 2.14	84.69 $\pm$ 0.71	79.37 $\pm$ 1.54	71.38
	HRCL(Our model)	89.17 $\pm$ 1.05	<u>83.23</u> $\pm$ 1.29	96.02 $\pm$ 1.07	89.59 $\pm$ 0.54	<u>85.26</u> $\pm$ 0.68	94.38 $\pm$ 0.58	84.36 $\pm$ 1.05	79.67
Re-TACRED	STAD*	81.60 $\pm$ 2.45	77.31 $\pm$ 2.32	86.39 $\pm$ 1.64	80.78 $\pm$ 1.72	82.43 $\pm$ 1.82	79.19 $\pm$ 2.31	75.48 $\pm$ 1.78	75.43
	fmLRE $\dagger$	<u>84.79</u> $\pm$ 1.54	80.29 $\pm$ 1.13	89.82 $\pm$ 0.89	<u>85.67</u> $\pm$ 0.85	86.41 $\pm$ 1.05	<u>84.94</u> $\pm$ 1.04	<u>78.29</u> $\pm$ 1.34	<u>79.37</u>
	MCCL $\dagger$	78.27 $\pm$ 2.73	73.24 $\pm$ 2.33	84.04 $\pm$ 1.79	75.39 $\pm$ 2.26	79.42 $\pm$ 2.06	71.75 $\pm$ 1.98	70.69 $\pm$ 2.37	70.39
	MatchPrompt $\dagger$	83.29 $\pm$ 1.45	<u>81.37</u> $\pm$ 1.50	85.30 $\pm$ 1.61	81.97 $\pm$ 1.74	80.64 $\pm$ 1.98	83.34 $\pm$ 2.45	76.29 $\pm$ 1.67	76.56
	GPT-Sen $\dagger$	79.50 $\pm$ 2.51	75.54 $\pm$ 1.91	83.89 $\pm$ 1.57	80.24 $\pm$ 2.37	81.69 $\pm$ 2.32	78.84 $\pm$ 2.59	71.54 $\pm$ 1.98	73.18
	HRCL(Our model)	86.27 $\pm$ 1.19	83.39 $\pm$ 0.78	<u>89.36</u> $\pm$ 0.59	87.45 $\pm$ 1.02	<u>85.78</u> $\pm$ 1.04	89.18 $\pm$ 1.67	78.46 $\pm$ 1.09	80.57

TABLE III

PERFORMANCE OF DIFFERENT MODELS IN TERMS OF B^3 , V-MEASURE, ARI, AND ACC (%) ON FEWREL AND SCIERC DATASETS. THE NUMBERS BEHIND THE “ $\pm$ ” SIGN REFER TO ITS STANDARD DEVIATION. METHODS MARKED WITH * ARE THE COPIED RESULTS DIRECTLY FROM THE ORIGINAL LITERATURE. METHODS MARKED WITH $\dagger$ REPRESENT THAT WE RE-IMPLEMENT THE MODELS FOR THIS TASK

Dataset	Models	B^3			V-measure			ARI	Acc(%)
		F_1	Prec.	Rec.	F_1	Hom.	Comp.		
FewRel	STAD $\dagger$	75.38 $\pm$ 1.79	76.18 $\pm$ 1.95	74.59 $\pm$ 1.67	73.19 $\pm$ 1.76	75.56 $\pm$ 2.37	70.96 $\pm$ 2.53	67.08 $\pm$ 2.09	70.14
	fmLRE $\dagger$	<u>81.27</u> $\pm$ 1.16	<u>80.29</u> $\pm$ 1.47	82.27 $\pm$ 2.46	<u>83.41</u> $\pm$ 1.22	80.19 $\pm$ 1.27	86.89 $\pm$ 1.46	<u>76.38</u> $\pm$ 1.45	<u>79.87</u>
	MCCL $\dagger$	79.89 $\pm$ 2.34	77.63 $\pm$ 2.02	82.29 $\pm$ 1.35	80.26 $\pm$ 2.04	<u>81.45</u> $\pm$ 1.96	79.10 $\pm$ 1.68	69.87 $\pm$ 1.75	73.56
	MatchPrompt*	79.23 $\pm$ 1.74	74.01 $\pm$ 1.93	<u>84.73</u> $\pm$ 2.02	77.64 $\pm$ 2.45	79.35 $\pm$ 2.05	76.00 $\pm$ 2.47	70.14 $\pm$ 1.69	71.28
	GPT-Sen $\dagger$	77.47 $\pm$ 2.05	78.21 $\pm$ 2.31	76.74 $\pm$ 1.32	73.29 $\pm$ 2.37	76.49 $\pm$ 1.89	70.35 $\pm$ 1.94	69.58 $\pm$ 1.76	75.47
	HRCL(Our model)	83.19 $\pm$ 1.03	80.34 $\pm$ 1.51	86.25 $\pm$ 1.46	85.36 $\pm$ 1.45	86.15 $\pm$ 1.20	<u>84.58</u> $\pm$ 1.57	79.49 $\pm$ 1.36	83.19
SciERC	STAD $\dagger$	50.18 $\pm$ 2.45	55.67 $\pm$ 1.96	45.67 $\pm$ 2.35	54.73 $\pm$ 1.87	50.74 $\pm$ 2.03	59.40 $\pm$ 2.67	44.38 $\pm$ 2.26	65.47
	fmLRE $\dagger$	<u>67.73</u> $\pm$ 1.69	65.49 $\pm$ 1.28	70.13 $\pm$ 1.34	<u>69.24</u> $\pm$ 0.63	<u>65.28</u> $\pm$ 1.47	<u>73.71</u> $\pm$ 1.23	<u>50.69</u> $\pm$ 1.49	<u>70.13</u>
	MCCL $\dagger$	63.74 $\pm$ 2.08	<u>67.13</u> $\pm$ 1.67	60.68 $\pm$ 1.46	65.38 $\pm$ 1.45	63.27 $\pm$ 1.54	67.64 $\pm$ 1.59	48.27 $\pm$ 1.54	67.69
	MatchPrompt $\dagger$	60.59 $\pm$ 2.16	63.24 $\pm$ 2.31	58.15 $\pm$ 1.78	62.79 $\pm$ 2.54	60.19 $\pm$ 2.09	65.62 $\pm$ 2.41	45.36 $\pm$ 1.78	62.35
	GPT-Sen $\dagger$	65.26 $\pm$ 2.57	64.29 $\pm$ 2.56	66.26 $\pm$ 1.94	64.57 $\pm$ 2.36	59.49 $\pm$ 2.32	70.60 $\pm$ 2.38	47.64 $\pm$ 1.59	65.68
	HRCL(Our model)	70.19 $\pm$ 1.47	73.24 $\pm$ 0.98	<u>67.38</u> $\pm$ 1.25	72.46 $\pm$ 1.19	70.14 $\pm$ 1.07	74.94 $\pm$ 0.97	54.37 $\pm$ 1.07	74.69

clustering relational categories. All results are averaged over the five independent testing runs.

B. Experimental Results and Analysis

Typically, clustering techniques usually limit the number of cluster centers in the baseline [9], [18], [24] to a maximum of 10. To ensure a fair comparison, we adjusted the p value in Algorithm 1 and set our final clustering class for the experiment to be 10. Although this is lower than the true relational classes existing in concentrated datasets, it still demonstrates HRCL’s capability to handle uneven label distribution.

1) *Main Results:* Tables II and III demonstrate the comparison performances of HRCL to all baselines in terms of B^3 , V-measure, ARI, and Acc (%) on five datasets. The optimal and suboptimal results are highlighted in boldface and underlined, respectively. Based on the statistical data from Tables II and III, we make the following observations.

1) fmLRE achieves the best performance among all the baselines on four datasets, making it the previous SOTA

method. However, MatchPrompt achieves the best performance on the TACRED dataset. We speculate the reason for this is that TACRED has a higher percentage (78.7%) of instances belonging to the “no_relation” type, which aligns better with the open RE model. HRCL consistently outperforms all baselines on B^3F_1 , V-measure F_1 , ARI, and Acc and average achieves 6.56% of performance improvement in B^3F_1 , 5.58% in V-measure F_1 , 5.34% higher in ARI, and 3.83% higher in Acc across five datasets when compared with fmLRE. Additionally, the standard deviation of HRCL is significantly lower than other baseline models. These results indicate that the proposed method has more vital generalization ability and robustness.

2) Compared to the contrastive learning clustering baselines (MCCL and MatchPrompt), there is a significant performance improvement on all datasets. This is because they solely focus on the clustering of external relational features while ignoring the hierarchical structure within the relational feature space. In contrast,

TABLE IV

PERFORMANCE IN TERMS OF B^3F_1 , V-MEASURE F_1 , AND ARI ON SEMEVAL AND RE-TACRED DATASETS WITH DIFFERENT DATA SAMPLE PROPORTION ($p\%$) FOR TRAINING AND VALIDATION SETS ON THE LOW-RESOURCE SETTING

Dataset	SemEval						Re-TACRED					
	B^3F_1			V-measure F_1			B^3F_1			V-measure F_1		
	5%	10%	30%	5%	10%	30%	5%	10%	30%	5%	10%	30%
STAD	67.36 $\pm$ 2.19	70.29 $\pm$ 2.05	79.34 $\pm$ 1.75	70.17 $\pm$ 2.12	75.28 $\pm$ 1.95	77.35 $\pm$ 1.27	73.68 $\pm$ 2.27	76.86 $\pm$ 1.75	79.47 $\pm$ 0.89	71.24 $\pm$ 2.27	75.34 $\pm$ 2.13	78.46 $\pm$ 1.56
fmLRE	75.34 $\pm$ 1.48	80.15 $\pm$ 1.87	86.38 $\pm$ 1.39	77.24 $\pm$ 2.57	81.18 $\pm$ 2.05	86.27 $\pm$ 1.03	75.39 $\pm$ 2.31	80.27 $\pm$ 1.64	84.65 $\pm$ 0.96	78.27 $\pm$ 2.31	81.24 $\pm$ 1.79	85.38 $\pm$ 1.06
MCCL	70.45 $\pm$ 2.53	75.73 $\pm$ 2.09	78.98 $\pm$ 1.56	72.56 $\pm$ 2.63	76.19 $\pm$ 1.04	78.92 $\pm$ 1.23	74.13 $\pm$ 2.54	76.89 $\pm$ 1.95	79.68 $\pm$ 1.53	74.57 $\pm$ 2.02	76.26 $\pm$ 1.74	79.98 $\pm$ 0.84
MatchPrompt	68.27 $\pm$ 2.36	73.19 $\pm$ 2.13	77.64 $\pm$ 2.47	71.67 $\pm$ 2.34	75.45 $\pm$ 1.56	78.13 $\pm$ 1.16	72.94 $\pm$ 2.57	75.62 $\pm$ 1.74	80.57 $\pm$ 1.26	73.47 $\pm$ 2.31	75.87 $\pm$ 1.97	78.64 $\pm$ 1.54
GPT-Sen	72.54 $\pm$ 2.41	76.79 $\pm$ 1.38	79.64 $\pm$ 1.03	73.24 $\pm$ 1.17	76.21 $\pm$ 1.67	78.54 $\pm$ 1.23	74.87 $\pm$ 2.24	77.19 $\pm$ 1.87	79.67 $\pm$ 1.58	73.56 $\pm$ 2.45	75.93 $\pm$ 1.78	79.78 $\pm$ 1.51
HRCL(Our model)	80.32$\pm$1.69	84.26$\pm$1.54	89.57$\pm$0.96	84.17$\pm$1.19	85.57$\pm$1.04	87.91$\pm$0.98	81.56$\pm$1.78	84.48$\pm$1.56	86.35$\pm$0.87	83.27$\pm$2.01	85.68$\pm$1.67	87.23$\pm$0.78

our model thoroughly explores hierarchical relationships with the same semantic information in a sentence. This suggests that exploring hierarchical clustering is of great significance in contrastive learning.

- 3) Compared to the traditional self-training approach (i.e., STAD) that relies on positive training instances, HRCL has achieved the greatest performance improvement. Its performance on TACRED showed a 28.78% improvement in terms of B^3F_1 . This performance gains, on the one hand, is due to higher-quality manually labeled samples in TACRED, and the other is attributed to the expressive power of HCA in delivering comprehensive interaction information between adjacent levels in propagation clustering, effectively avoiding error accumulation.
- 4) Compared to the GPT-3 baseline, HRCL shows a 5%–12% improvement in performance for B^3F_1 and V-measure F_1 . The improvement can be attributed to two factors: a) incorporating pre-defined structural information, such as instruction description, relation, and entity type, as prompts enhance semantic understanding for the PLM encoder and b) since the GPT-3 series model is not open-source, we are unable to fine-tune corresponding parameters or determine if the model has been trained on the datasets used in the low-resource setting. Therefore, we believe that this significant difference in results may be due to these reasons. This demonstrates the effectiveness of our designed prompt information during the encoding phase.

2) *Results With Random Data Samples:* To further investigate the stability of our proposed method, we conducted additional experiments on SemEval and Re-TACRED datasets, which have a larger number of instances. For each experiment, we randomly sample a fixed proportion ($p\%$) from the training and validation sets for the low-resource setting and report the average values of B^3F_1 , V-measure F_1 , and ARI, as well as their standard deviation across five runs using different random seeds. Results are represented in Table IV.

From Table IV, we can observe that HRCL consistently outperforms all the baselines. More specifically, when compared to the previous SOTA model fmLRE, HRCL achieves an average improvement of 4.09% and 4.03% in terms of B^3F_1 on SemEval and Re-TACRED, respectively, across different sample proportions. Additionally, HRCL exhibits greater stability than other baselines when considering standard deviation. Considering the extremely low-resource scenario, e.g., 5% sample proportion for SemEval and Re-TACRED, HRCL achieves an average 5.58% B^3F_1 boost compared with fmLRE.

While further increasing the number of labeled samples to 10%–30%, the average improvement in B^3F_1 decreases to 4.16% and 2.45% on these two datasets, respectively. This consistent improvement indicates the effectiveness of the explicit feedback mechanism in relation-wise contrastive learning. Furthermore, with the increase of ratio in sample data, the performance improvement of HRCL tends to stabilize, which also demonstrates the adaptability of our proposed model in the low-resource setting.

3) *Ablation Studies:* To further investigate the effectiveness of different components in HRCL, we perform the ablation study on SemEval, TACRED, and FewRel datasets and report results in Table V. Specifically, w/o TP indicates removing the Task-related Prompt in input text during the encoding phase; w/o HiPC using k -nearest neighbors (kNN) and k -means instead of the hierarchical propagation clustering and report the best performance between them; w/o HCA represents removing the hierarchy cross-attention modules in HiPC Process; w/o ReNCE represents the models only with InfoNCE loss to retain the local smoothness during the clustering process. From Table V, we observe the following holds.

- 1) Without task-related knowledge to guide PLM in extracting relational features (w/o TP), the performance decreases by 2%–5% in terms of B^3/V -measure F_1 on SemEval, TACRED, and FewRel, respectively. This result suggests that instruction and schema-constrained information are indeed beneficial for feature extraction during the encoding phase.
- 2) Using kNN and k -means instead of the HiPC (w/o HiPC) during the clustering phase, the performance drops are 4%–10%, 5%–9%, and 4%–7%, which demonstrate the effectiveness of the proposed model for LRE by clustering the relational features with APC in a hierarchical structure.
- 3) HCA transfers relational features from high-level relations to base-level through hierarchy cross-attention in the clustering process. Removing the HCA module (HRCL_{w/o} HCA) significantly decreases the performance of HRCL, indicating that this component provides high-quality signals for training a better relational cluster model. Furthermore, Fig. 4 shows ReNCE had lower loss and faster convergence than w/o HCA.
- 4) The main purpose of HiNCE is to leverage relation-wise contrastive learning in addition to instance-wise. Without ReNCE and only leaving the InfoNCE alone (w/o ReNCE), the RE performance average drops 6.49% in terms of B^3F_1 over three datasets, which decrease is the sharpest except for the w/o HiPC. This result

TABLE V

QUANTITATIVE RESULTS (%) WITH RESPECT TO B^3F_1 , V-MEASURE F_1 , AND ARI (MEAN $\pm$ STANDARD DEVIATION) ON SEMEVAL, TACRED, AND FEWREL DATASETS. Δ REPRESENTS THE GAIN OF VARIANTS COMPARED WITH HRCL

Models	SemEval			TACRED			FewRel		
	B^3F_1	V-measure F_1	ARI	B^3F_1	V-measure F_1	ARI	B^3F_1	V-measure F_1	ARI
HRCL _{w/oTP} (Δ)	85.24 $\pm$ 2.17 ($\downarrow$ 4.23)	84.25 $\pm$ 1.68 ($\downarrow$ 2.88)	67.47 $\pm$ 1.87 ($\downarrow$ 0.92)	86.53 $\pm$ 2.34 ($\downarrow$ 2.64)	85.42 $\pm$ 4.17 ($\downarrow$ 4.91)	82.28 $\pm$ 1.73 ($\downarrow$ 2.08)	80.27 $\pm$ 2.41 ($\downarrow$ 2.92)	82.49 $\pm$ 2.30 ($\downarrow$ 2.87)	76.18 $\pm$ 1.36 ($\downarrow$ 3.31)
HRCL _{w/oHiPC} (Δ)	80.15 $\pm$ 2.46 ($\downarrow$ 9.32)	80.36 $\pm$ 2.54 ($\downarrow$ 6.77)	63.46 $\pm$ 1.97 ($\downarrow$ 4.93)	83.42 $\pm$ 2.05 ($\downarrow$ 5.75)	81.26 $\pm$ 1.98 ($\downarrow$ 8.33)	80.45 $\pm$ 1.76 ($\downarrow$ 3.91)	78.56 $\pm$ 1.39 ($\downarrow$ 4.63)	79.35 $\pm$ 2.31 ($\downarrow$ 6.01)	73.58 $\pm$ 2.67 ($\downarrow$ 5.91)
HRCL _{w/oHCA} (Δ)	82.34 $\pm$ 2.07 ($\downarrow$ 7.13)	81.57 $\pm$ 2.18 ($\downarrow$ 5.56)	65.89 $\pm$ 1.76 ($\downarrow$ 2.50)	84.78 $\pm$ 2.46 ($\downarrow$ 4.39)	83.41 $\pm$ 1.79 ($\downarrow$ 6.18)	82.32 $\pm$ 1.87 ($\downarrow$ 2.04)	78.89 $\pm$ 1.81 ($\downarrow$ 4.30)	80.17 $\pm$ 1.93 ($\downarrow$ 5.19)	74.39 $\pm$ 1.85 ($\downarrow$ 5.10)
HRCL _{w/oReNCE} (Δ)	81.21 $\pm$ 2.17 ($\downarrow$ 8.26)	81.03 $\pm$ 1.88 ($\downarrow$ 6.10)	64.57 $\pm$ 2.04 ($\downarrow$ 3.82)	82.46 $\pm$ 1.76 ($\downarrow$ 6.71)	82.28 $\pm$ 1.80 ($\downarrow$ 7.31)	81.67 $\pm$ 2.04 ($\downarrow$ 2.69)	78.69 $\pm$ 1.91 ($\downarrow$ 4.50)	79.61 $\pm$ 2.15 ($\downarrow$ 5.75)	74.02 $\pm$ 1.83 ($\downarrow$ 5.47)

reflects the effectiveness of ReNCE, indicating the model enables obtaining better representations to form more semantically meaningful clusters through the feedback mechanism of ReNCE.

Furthermore, we visualize the hierarchical contextual relation features $\mathbb{R}^{(2+v+p) \times d_R}$ after dimension reduction using t-SNE [41] to more intuitively show the hierarchical structure clustering. Specifically, we randomly selected 10 000 relation instances from the TACRED dataset and presented the visualization results for different settings in Fig. 5. Fig. 5(a) shows a highly accurate clustering outcome achieved by utilizing the HRCL algorithm to extract high-level relational features, where features are colored according to their clustering labels. In Fig. 5(b)–(d), we give the base-level clustering relation labels with different shapes while reserving the color series of the corresponding high-level relation category. This allows us to explore how different modules utilize high-level relation features to guide the clustering of base-level relations.

Experimental results show that removing HCA from HRCL in Fig. 5(b) does not significantly affect the clustering results for denser clusters. However, ReNCE is removed from HRCL in Fig. 5(c), and it becomes difficult to attain high-quality relational semantics without the assistance of relation-wise knowledge during the process of exemplar points update, which leads to loose, even incorrect clustering results. When the kNN algorithm is used to perform clustering on the high-level relational features in Fig. 5(d), it results in poor clustering and confusion between similar points due to the inability to utilize hierarchical information to update relational features or mutual similarity between feature points. These outcomes demonstrate that HRCL can effectively leverage relational features to clustering in low-resource settings and distinguish similar and dissimilar instances.

- 5) We also report the standard deviations of each HRCL version, and results show that all HRCL variants have relatively stable performances with low fluctuation, which verifies that performance degradation between the complete model is due to removing specific components rather than experimental error.

4) *Hyperparameter Sensitivity Analysis:* This section investigates the impact of several vital hyper-parameters on the performance of LRE.

- 1) *Analysis of the number of random span words p :* The value of p will affect the fixed length of the relational representation in (2). We report the average performance of B^3F_1 , V-measure F_1 , and ARI on SemEval and TACRED datasets by varying the p from 1 to 4 ($\leq 1, 2, 3, 4$ and ≥ 5). The flux results in Table VI demonstrate that both knowledge missing and redundant in relational representation will hurt the model’s performance. Short-span words provide insufficient information, making it difficult to cover large-scale sentence representations. On the other hand, long-span words increase computational complexity and result in information redundancy, also decreasing model performance. To further confirm our conclusion, we re-implemented the top-three models from the main results using various p values. The experimental results depicted in Fig. 6 align with the initial conclusions mentioned above.
- 2) *Analysis on the level of K hierarchical layers:* We study the level of K hierarchical layers and how to get the preference sequences to discover the most appropriate tree-structured hierarchical relation for the data distribution. Specifically, the experimental results are reported in Table VI by varying the K values from 2 to 5 ($\leq 2, 3, 4, 5$ and $\geq 3 + V$) with fixed preference p_u . The unstable performance indicates that fewer layers fail to exchange information well, while more layers can cause instance-level information confusion between different levels of granularity layers. The study [42] has demonstrated that the diagonal, minimum, and median values of the similarity matrix are the best preferences for propagation clustering, so we manually adjust the preference sequence between them multiple times with $K = 4$ and obtain the average performance as $3+V$ to compare with the automatically generated ones by Line 1 in HiPC. Furthermore, the experiment result indicates that the preference p_u to take the median values of the similarity matrix is practicability and effectiveness for the model’s performance. Thus, we take the median values of s_{ij} as the basis for calculating the final preference p_γ .

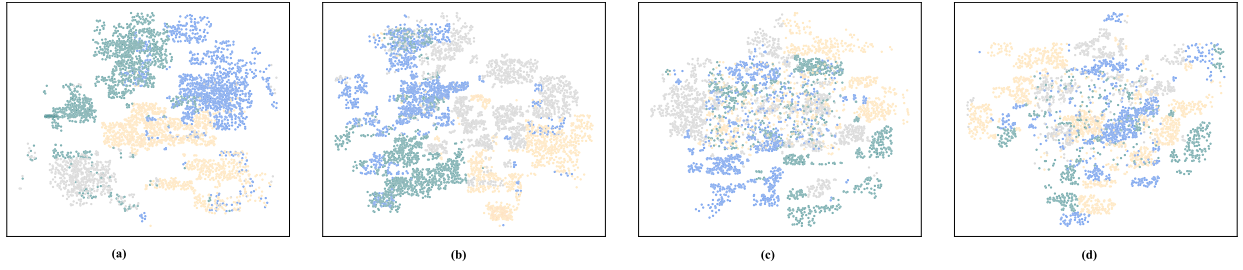


Fig. 5. Visualization of relational features with t-SNE dimension reduction on the TACRED dataset. Each point presents one sample point. Different colors used in the figure correspond to their relational labels. (a) HRCL (high-level relational features). (b) HRCL w/o HCA. (c) HRCL w/o ReNCE. (d) HRCL w/o HiPC.

TABLE VI
AVERAGE PERFORMANCE WITH DIFFERENT VARIABLE ON SEMEval AND TACRED DATASETS. AP REPRESENTS THE AVERAGE PERFORMANCE OF B_3 F1, V-MEASURE F1, AND ARI

Dataset	SemEval					TACRED				
Variable	1	2	3	4	5	1	2	3	4	5
Random Span(p)	1	2	3	4	5	1	2	3	4	5
AP	79.17 $\pm$ 1.95	80.28 $\pm$ 1.56	81.69 $\pm$ 1.01	80.35 $\pm$ 1.57	78.16 $\pm$ 1.79	84.37 $\pm$ 1.74	85.64 $\pm$ 1.98	87.73 $\pm$ 1.13	86.54 $\pm$ 1.78	83.71 $\pm$ 1.95
K layers	2	3	4	5	3+V	2	3	4	5	3+V
AP	77.64 $\pm$ 2.03	81.16 $\pm$ 1.73	81.74 $\pm$ 1.23	80.28 $\pm$ 2.15	81.59 $\pm$ 1.92	81.25 $\pm$ 2.35	83.24 $\pm$ 1.79	87.69 $\pm$ 1.83	86.17 $\pm$ 1.86	87.91 $\pm$ 2.05
p_u	Median(s_{ij})	Mean(s_{ij})	Digonal(s_{ij})	-	-	Median(s_{ij})	Mean(s_{ij})	Digonal(s_{ij})	-	-
AP	81.71 $\pm$ 2.03	80.23 $\pm$ 1.38	78.86 $\pm$ 1.89	-	-	87.53 $\pm$ 1.43	84.32 $\pm$ 2.18	80.16 $\pm$ 1.56	-	-

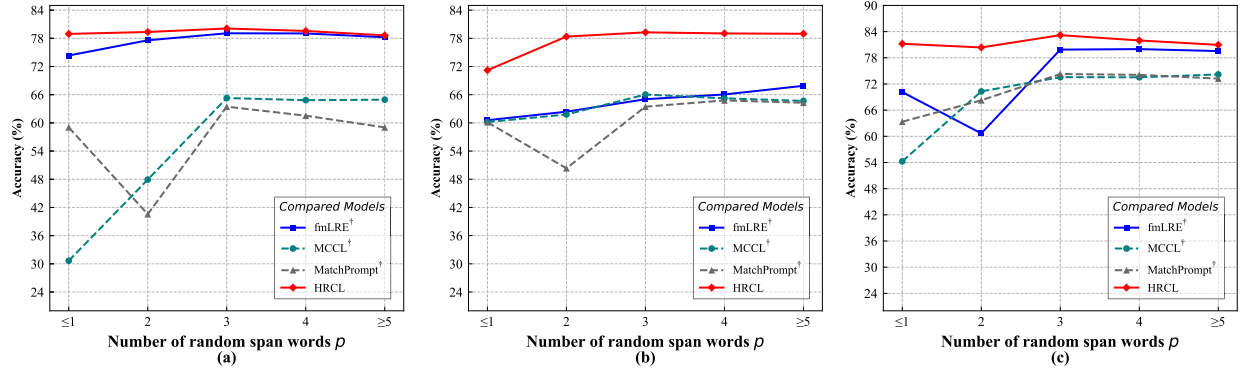


Fig. 6. Performance of the top-three baselines and the proposed model with varying random span words p (≤ 1 , 2, 3, 4 and ≥ 5) on (a) SemEval, (b) TACRED, and (c) FewRel datasets. $\dagger$ denotes the re-implemented models with varying p values.

3) *Analysis of computing resource*: Table VII presents a comparison of computing resources of HRCL and three representative baselines, i.e., fmLRE, MatchPrompt, and MCCL. Among them, fmLRE is a self-training method that conducts full-batch training iteratively, while MatchPrompt, MCCL, and our model are feature clustering models and can be utilized with small-batch strategy. From Table VII, we clearly observe that clustering models have fewer parameters and with less training time, but the inference time increased. The main reason is because the contrastive learning model divides the hypermeters into various batches and only consumes per-batch memory when training. In addition, the reasoning stage is also a clustering process, so the inference time is significantly longer than the fmLRE baseline.

5) *Case Study*: We compare HRCL with other baselines through case studies. Due to space limitations, we only compare HRCL with fmLRE and HRCLw/o HiPC. The module instance is consistent with the setting in the ablation study. Additionally, when the model makes a wrong prediction, it tends to be predicted as unknown, which is similar to what happens in module comparison. Three examples are presented in the two comparisons and shown in Table VIII. First, in the

TABLE VII

COMPARISON OF VARIOUS COMPUTING RESOURCES. Para_{all} IS THE NUMBER OF PARAMETERS IN THE WHOLE MODEL. Para_{tuned} DENOTES THE TUNE PROPORTION OF ALL PARAMETERS. EPO_TIME DENOTES THE EPOCH TIME TAKING THE AVERAGE TIME (HOUR) OF THE MODEL TO FINISH ONE EPOCH. INTIME DENOTES THE INFERENCE TIME FOR EACH SAMPLE (SEC/SAMPLE)

Model	Para _{all}	Para _{tuned}	Epo_time (h)	InTime
fmLRE $\dagger$	121,045,289	6.18%	4.68	3.8
MCCL $\dagger$	110,873,299	5.38%	2.21	6.1
MatchPrompt $\dagger$	116,389,421	7.59%	3.59	5.7
HRCL	112,537,675	6.71%	3.71	4.2

model comparison, there is no significant difference among the performances of each model of these cases. However, HRCL demonstrates greater stability and accurately identifies all entities and relations. Second, fmLRE incorrectly recognizes all entities and relations in the first and second instances due to insufficient useful relational features during self-training. Conversely, HRCL w/o HiPC also misidentifies the relational types of *city_of_headquarters* and *date_of_birth* as *member_of* and *shareholders* in the second and third instances because it cannot cluster valuable semantic knowledge without hierarchical relation-wise clustering.

TABLE VIII

EXTRACTION OF RELATION CATEGORY WITH DIFFERENT MODELS. THE HEAD ENTITY AND TAIL ENTITY ARE MARKED WITH BLUE, AND THE RELATIONAL CATEGORY IS MARKED WITH RED. THE MISIDENTIFIED RELATION CATEGORIES ARE MARKED WITH ✕ SYMBOL AND MAY APPEAR SIMILAR TO INTERFERENCE ENTITIES MARKED WITH GREEN

Input Sentences	Ground-Truth	fmLRE	HRCL w/o HiPC	HRCL
Sharpton has said he will not endorse any candidate until hearing more about their views on civil rights and other issues at his National Action Network convention next week in New York City.	<H: Sharpton > <T: National Action Network > <R: founded_by >	<H: Candidate > <T: New York City > <R: cities_of_birth >✕	<H: Sharpton > <T: National Action Network > <R: founded_by >✓	<H: Sharpton > <T: National Action Network > <R: founded_by >
In a statement, Miami, Florida-based Carnival Cruise Lines said the Wednesday incident occurred during an outing at a Limon beach that a group of a dozen passengers had arranged on their own.	<H: Carnival Cruise Lines > <T: Miami > <R: city_of_headquarters >	<H: Wednesday incident > <T: a dozen passengers > <R: cause_of_death >✕	<H: a dozen passengers > <T: Wednesday incident > <R: member_of >✕	<H: Carnival Cruise Lines > <T: Miami > <R: city_of_headquarters >
Born on Aug. 15, 1925, in a poor neighborhood of Montreal, Peterson got his passion for music from his father.	<H: Peterson > <T: Aug. 15, 1925 > <R: date_of_birth >	<H: Peterson > <T: Aug. 15, 1925 > <R: date_of_birth >✓	<H: Peterson > <T: passion for music > <R: shareholders >✕	<H: Peterson > <T: Aug. 15, 1925 > <R: date_of_birth >

V. CONCLUSION

In this article, we propose a novel contrastive learning framework called HRCL for LRE. We adopt task-based prompts to enhance relational representations for input sentences. HRCL leverages HiPC to aggregate hierarchical exemplar points with high-level relation features, which differs from traditional self-training methods that endure gradual drift and instance-wise contrastive learning models that confuse relational semantics. In turn, these clustered exemplars are further utilized to update the relational features of each sentence iteratively and are optimized by conducting instance and relation-wise contrastive learning through the designed HiNCE module. Extensive experiments on five public datasets have demonstrated the effectiveness of HRCL compared with current SOTA models.

For future work, we aim to delve into more advanced contrastive learning algorithms to achieve more precise relation clustering between instances in low-resource settings. Furthermore, we will explore more efficient clustering and data augmentation methods tailored explicitly for contrastive learning. Another aspect of our future work involves evaluating the performance of HRCL on similar and adversarial LRE datasets. We aspire that our research can offer a novel perspective for further LRE studies.

REFERENCES

- [1] A. Hur, N. Janjua, and M. Ahmed, "A survey on state-of-the-art techniques for knowledge graphs construction and challenges ahead," in *Proc. IEEE 4th Int. Conf. Artif. Intell. Knowl. Eng. (AIKE)*, Dec. 2021, pp. 99–103.
- [2] F. Corcoglioniti, M. Dragoni, M. Rospocher, and A. P. Aprosio, "Knowledge extraction for information retrieval," in *Proc. Eur. Semantic Web Conf.* Berlin, Germany: Springer, 2016, pp. 317–333.
- [3] X. Han et al., "FewRel: A large-scale supervised few-shot relation classification dataset with state-of-the-art evaluation," in *Proc. Conf. Empirical Methods Natural Language Process.*, 2018, pp. 4803–4809.
- [4] H. Peng et al., "Learning from context or names? An empirical study on neural relation extraction," in *Proc. Conf. Empirical Methods Natural Language Process. (EMNLP)*, 2020, pp. 3661–3672.
- [5] Y. Qin et al., "ERICA: Improving entity and relation understanding for pre-trained language models via contrastive learning," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics, 11th Int. Joint Conf. Natural Lang. Process.*, 2021, pp. 3350–3363.
- [6] M. Mintz, S. Bills, R. Snow, and D. Jurafsky, "Distant supervision for relation extraction without labeled data," in *Proc. Joint Conf. 47th Annu. Meeting ACL, 4th Int. Joint Conf. Natural Lang. Process.*, 2009, pp. 1003–1011.
- [7] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.
- [8] M. Dong, C. Pan, and Z. Luo, "MapRE: An effective semantic mapping approach for low-resource relation extraction," 2021, *arXiv:2109.04108*.
- [9] W. Zhou, S. Zhang, T. Naumann, M. Chen, and H. Poon, "Continual contrastive finetuning improves low-resource relation extraction," 2022, *arXiv:2212.10823*.
- [10] C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals, "Understanding deep learning (still) requires rethinking generalization," *Commun. ACM*, vol. 64, no. 3, pp. 107–115, 2021.
- [11] X. Hu, S. Liu, C. Zhang, S. Li, L. Wen, and P. S. Yu, "HiURE: Hierarchical exemplar contrastive learning for unsupervised relation extraction," 2022, *arXiv:2205.02225*.
- [12] L. B. Soares, N. FitzGerald, J. Ling, and T. Kwiatkowski, "Matching the blanks: Distributional similarity for relation learning," in *Proc. 57th Annu. Meeting Assoc. Comput. Linguistics*, 2019, pp. 2895–2905.
- [13] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol.*, vol. 1, Jun. 2019, pp. 4171–4186.
- [14] W. Li and T. Qian, "Exploit multiple reference graphs for semi-supervised relation extraction," 2020, *arXiv:2010.11383*.
- [15] X. Hu, C. Zhang, F. Ma, C. Liu, L. Wen, and P. S. Yu, "Semi-supervised relation extraction via incremental meta self-training," in *Proc. Findings Assoc. Comput. Linguistics*, 2021, pp. 487–496.
- [16] N. Zhang et al., "Long-tail relation extraction via knowledge graph embeddings and graph convolution networks," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics*, 2019, pp. 3016–3025.
- [17] X. Hu et al., "Gradient imitation reinforcement learning for low resource relation extraction," 2021, *arXiv:2109.06415*.
- [18] J. Yu, X. Wang, J. Zhao, C. Yang, and W. Chen, "STAD: Self-training with ambiguous data for low-resource relation extraction," 2022, *arXiv:2209.01431*.
- [19] L. Jiang, Z. Zhou, T. Leung, L.-J. Li, and L. Fei-Fei, "MentorNet: Learning data-driven curriculum for very deep neural networks on corrupted labels," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 2304–2313.
- [20] X. Hu, L. Wen, Y. Xu, C. Zhang, and P. Yu, "SelfORE: Self-supervised relational feature learning for open relation extraction," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2020, pp. 3673–3682.
- [21] D. Li, T. Zhang, N. Hu, C. Wang, and X. He, "HiCLRE: A hierarchical contrastive learning framework for distantly supervised relation extraction," 2022, *arXiv:2202.13352*.
- [22] M. Li et al., "Keywords and instances: A hierarchical contrastive learning framework unifying hybrid granularities for text generation," 2022, *arXiv:2205.13346*.
- [23] K. Zhao, H. Xu, J. Yang, and K. Gao, "Consistent representation learning for continual relation extraction," 2022, *arXiv:2203.02721*.
- [24] S. Wang, B. Zhang, Y. Xu, Y. Wu, and B. Xiao, "RCL: Relation contrastive learning for zero-shot relation extraction," in *Proc. Findings Assoc. Comput. Linguistics, (NAACL)*, 2022, pp. 2456–2468.
- [25] B. J. Frey and D. Dueck, "Clustering by passing messages between data points," *Science*, vol. 315, no. 5814, pp. 972–976, Feb. 2007.
- [26] D. Gribel, M. Gendreau, and T. Vidal, "Semi-supervised clustering with inaccurate pairwise annotations," *Inf. Sci.*, vol. 607, pp. 441–457, Aug. 2022.
- [27] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.
- [28] I. Hendrickx et al., "SemEval-2010 Task 8: Multi-way classification of semantic relations between pairs of nominals," in *Proc. 5th Int. Workshop Semantic Eval.*, 2010, pp. 33–38.

- [29] L. T. Yu, W. N. Zhang, J. Wang, and Y. Yu, "SeqGAN: Sequence generative adversarial nets with policy gradient," in *Proc. AAAI*, vol. 31, 2017, pp. 2852–2858.
- [30] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. NeurIPS*, 2017, pp. 1195–1204.
- [31] H. T. Lin, J. Yan, M. Qu, and X. Ren, "Learning dual retrieval module for semi-supervised relation extraction," in *Proc. WWW*, 2019, pp. 1073–1083.
- [32] Y. Zhang, V. Zhong, D. Chen, G. Angeli, and C. D. Manning, "Position-aware attention and supervised data improve slot filling," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2017, pp. 35–45.
- [33] Y. Yu, X. Si, C. Hu, and J. Zhang, "A review of recurrent neural networks: LSTM cells and network architectures," *Neural Comput.*, vol. 31, no. 7, pp. 1235–1270, Jul. 2019.
- [34] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 4080–4090.
- [35] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 8024–8035.
- [36] L. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–19.
- [37] É. Simon, V. Guigue, and B. Piwowarski, "Unsupervised information extraction: Regularizing discriminative approaches with relation distribution losses," in *Proc. 57th Annu. Meeting Assoc. Comput. Linguistics*, Florence, Italy, New York, NY, USA: Association for Computational Linguistics, 2019, pp. 1378–1387.
- [38] A. Bagga and B. Baldwin, "Entity-based cross-document coreferencing using the vector space model," in *Proc. 17th Int. Conf. Comput. Linguistics*, 1998, pp. 79–85.
- [39] A. Rosenberg and J. Hirschberg, "V-measure: A conditional entropy-based external cluster evaluation measure," in *Proc. EMNLP-CoNLL*, 2007, pp. 410–420.
- [40] L. Hubert and P. Arabie, "Comparing partitions," *J. Classification*, vol. 2, no. 1, pp. 193–218, Dec. 1985.
- [41] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, Nov. 2008.
- [42] A. F. Moiane and Á. M. L. Machado, "Evaluation of the clustering performance of affinity propagation algorithm considering the influence of preference parameter and damping factor," *Boletim de Ciências Geodésicas*, vol. 24, no. 4, pp. 426–441, Dec. 2018.
- [43] Y. Luan, L. He, M. Ostendorf, and H. Hajishirzi, "Multi-task identification of entities, relations, and coreference for scientific knowledge graph construction," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2018, pp. 3219–3232.
- [44] G. Stoica, E. A. Platanios, and B. Póczos, "Re-TACRED: Addressing shortcomings of the TACRED dataset," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 15, pp. 13843–13850.
- [45] P. Wang, T. Shao, K. Ji, G. Li, and W. Ke, "fmLRE: A low-resource relation extraction model based on feature mapping similarity calculation," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 13700–13708.
- [46] J. Wang, L. Zhang, J. Liu, X. Liang, Y. Zhong, and Y. Wu, "MatchPrompt: Prompt-based open relation extraction with semantic consistency guided clustering," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2022, pp. 7875–7888.
- [47] T. Gao, X. Yao, and D. Chen, "SimCSE: Simple contrastive learning of sentence embeddings," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2021, pp. 6894–6910.
- [48] L. Ouyang et al., "Training language models to follow instructions with human feedback," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 27730–27744.
- [49] H. Won Chung et al., "Scaling instruction-finetuned language models," 2022, *arXiv:2210.11416*.
- [50] C. Rosenberg, M. Hebert, and H. Schneiderman, "Semi-supervised self-training of object detection models," in *Proc. 7th IEEE Workshops Appl. Comput. Vis.*, Jan. 2005.
- [51] H. Ye et al., "Ontology-enhanced prompt-tuning for few-shot learning," in *Proc. ACM Web Conf.*, Apr. 2022, pp. 778–787.
- [52] X. Chen et al., "KnowPrompt: Knowledge-aware prompt-tuning with synergistic optimization for relation extraction," in *Proc. ACM Web Conf.*, Apr. 2022, pp. 2778–2788.
- [53] X. Chen, H. Fan, R. Girshick, and K. He, "Improved baselines with momentum contrastive learning," 2020, *arXiv:2003.04297*.
- [54] O. Levy, M. Seo, E. Choi, and L. Zettlemoyer, "Zero-shot relation extraction via reading comprehension," in *Proc. 21st Conf. Comput. Natural Lang. Learn. (CoNLL)*, 2017, pp. 333–342.
- [55] S. Deng, Y. Ma, N. Zhang, Y. Cao, and B. Hooi, "Information extraction in low-resource scenarios: Survey and perspective," 2022, *arXiv:2202.08063*.
- [56] J. Yu, T. Zhu, W. Chen, W. Zhang, and M. Zhang, "Improving relation extraction with relational paraphrase sentences," in *Proc. 28th Int. Conf. Comput. Linguistics*, 2020, pp. 1687–1698.
- [57] X. Zhan, J. Xie, Z. Liu, Y.-S. Ong, and C. C. Loy, "Online deep clustering for unsupervised representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6687–6696.
- [58] S. Li, X. Hu, L. Lin, A. Liu, L. Wen, and P. S. Yu, "A multi-level supervised contrastive learning framework for low-resource natural language inference," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 31, pp. 1771–1783, 2023.
- [59] X. Peng, J. Cheng, X. Tang, J. Liu, and J. Wu, "Dual contrastive learning network for graph clustering," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–11, 2023.
- [60] W. Tu, S. Zhou, X. Liu, C. Ge, Z. Cai, and Y. Liu, "Hierarchically contrastive hard sample mining for graph self-supervised pretraining," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–14, 2024.
- [61] C. Yang, Y.-M. Cheung, J. Ding, K. C. Tan, B. Xue, and M. Zhang, "Contrastive learning assisted-alignment for partial domain adaptation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 7621–7634, 2022.
- [62] O. Maddouri, X. Qian, and B.-J. Yoon, "Geometric affinity propagation for clustering with network knowledge," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 11, pp. 11419–11436, 2023.
- [63] J. Liu, G. Liao, J. Xu, S. Zhu, C. Zeng, and F. H. Juwono, "Unsupervised affinity propagation clustering based clutter suppression and target detection algorithm for non-side-looking airborne radar," *Remote Sens.*, vol. 15, no. 8, p. 2077, Apr. 2023.



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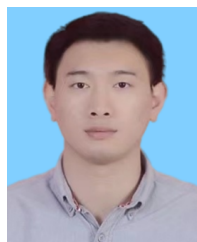
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